

FIN Local Research — Release 10.46

Checkpoint P475–P480–P484

Dependency-Minimal Exact Replay, Algebraic Elimination Burden,
and an Adversarial Phase-Face Audit

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Durable result. The exact O180/O181 rational Krawczyk and positivity certificate can now be independently replayed with the Python standard library alone. A tampered certificate is rejected.

Adversarial correction. The proposed exact complex phase-face theorem contains one unpaid implication: the invariant three-dimensional axis has not been proved to satisfy the full causal equal-endpoint constraint. An exact counterexample shows that odd dimension and active Riccati congruence alone are insufficient. Full-cone nonuniqueness therefore remains strong evidence, not a theorem.

Algebraic boundary. Exact lexicographic elimination did not produce a polynomial for the optimum inside a 45-second, 3-GiB local envelope. This is a method-specific resource result, not mathematical impossibility.

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Publication statement

This monograph reports local analytical and computational Programs P475, P480, and P484. It uses no network, Firecrawl, external AI, external audit, remote computation, laboratory record, or empirical physical evidence.

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Chapter 1

Executive summary

This checkpoint follows the exact O167 attainment certificate of P473 and its uniform local continuation in P483. It was designed to do three different jobs: reduce verifier dependencies, attack the algebraic value directly, and attempt to turn the P474 complex-flat-direction evidence into a theorem.

1. **P480** — **[Computer-assisted proof] replayed**. A serialized rational document contains the 13 equations, all 169 Jacobian entries, the rational root box, algebraic sine enclosures, preconditioner, center matrices, and positivity bounds. A new checker using only the Python standard library recomputes exact interval Krawczyk inclusion and exact Sylvester payments. It rejects a zeroed preconditioner in an adversarial test.
2. **P484** — **[Strong evidence]; exact step open**. Nine exact symbolic identities yield a clean three-plus-three parity reduction. Six numerical tangent equations agree through 80 displayed digits, and residuals are at machine precision. The attempted analytic proof nevertheless needs the fixed covector of a metric rotation to lie in a particular causal plane. An exact rational counterexample proves that odd-dimensional metric orthogonality and the active Riccati equations do not imply this. The repository-specific shared-entry and standard-sector identities may still imply it, but that claim has not been certified. The exact phase face and full-cone nonuniqueness remain open.
3. **P475** — **[Open]; resource-bounded no-go**. All sine coefficients reduce to the degree-four field $\mathbb{Q}(\sin(\pi/8))$. The exact Riccati system is cubic over that field and affine in the three normalizer coordinates. A bounded exact lexicographic Gröbner attempt returned no univariate relation for the value L before its hard resource stop. This does not imply that no minimal polynomial exists.

The strongest positive result is the dependency-minimal P480 replay. The strongest falsification result is the P484 correction: a beautiful structural argument is not promoted beyond the exact implication actually paid by the artifacts.

The new objects are O185 Conditional Odd-Dimensional Complex Comb Phase Face, O186 Dependency-Minimal Exact O181 Replay Certificate, and O187 Resource-Bounded Algebraic Elimination Burden Certificate.

Chapter 2

Confidence convention and state transition

Label	Meaning
[Proven]	Exact analytic or finite symbolic consequence of declared premises
[Computer-assisted proof]	Finite rational or outward-interval certificate with auditable artifacts
[Strong evidence]	Reproducible numerical evidence missing one exact certificate
[Conditional]	Exact implication after an explicitly named open premise
[Open]	Current local artifacts do not decide the proposition
[Refuted]	A necessary implication fails by exact counterexample

Lane	Status	Boundary
Nominal O181 root and O167 attainment	[Computer-assisted proof]	Declared reduced ordered three-slot comb
P483 parameter tube	[Computer-assisted proof]	Tiny dimensionless rectangle; one local box
Dependency-minimal replay	[Computer-assisted proof] replayed	Finite root and positivity part only
Full complex optimizer uniqueness	[Open]	Strong evidence against; exact causal-axis step missing
Minimal polynomial of L	[Open]	Blind exact lexicographic route exceeded declared resources
Lean formalization	Source only	No local Lean toolchain; not machine checked
Selector / QW-2191	[Open]	No non-premise strict source
Dimensional source	Missing	No time, length, action, mass, or energy unit
Physical realization	Externally blocked	No apparatus, calibration, custody, or laboratory record
Legacy-strict bridge and role transfer	Incomplete	No silent substitution or inherited role

Chapter 3

Inherited exact operator problem

For the supplied compressed Hermitian process difference Δ , the declared ordered three-slot primal optimizes

$$\frac{1}{2} \text{Tr}[\Delta(T_+ - T_-)]$$

over positive tester effects with the nested comb normalization. At a fixed positive normalizer $N = T_+ + T_-$, the inner value is

$$F(N) = \frac{1}{2} \left\| N^{1/2} \Delta N^{1/2} \right\|_1.$$

P473 certified an exact positive root of the structured Riccati system

$$X_3 N X_3 = \frac{1}{4} \Delta N \Delta \tag{3.1}$$

inside a rational 13-dimensional box of radius 3×10^{-14} . The block pattern pays all lower dual equalities; the positive support theorem and trace telescope then give exact primal–dual attainment. P483 proved the same conclusion uniformly for

$$|q - 4/5| \leq 3 \times 10^{-9}, \quad |\theta - \pi/8| \leq 3 \times 10^{-9}.$$

These are local root results in a real polynomial chart. They do not establish uniqueness in the complete complex causal cone.

Chapter 4

P480: dependency-minimal exact replay

4.1 Question and success criterion

The exact numerical certificate previously depended on a builder using NumPy, SciPy, and SymPy. P480 asks whether a verifier can recompute the finite proof using only exact rational arithmetic available in a default Python installation. Success requires recomputation rather than trust in stored Boolean fields.

4.2 Certificate schema

The JSON certificate serializes:

- 13 rational polynomial equations as monomial/exponent lists;
- all $13 \times 13 = 169$ rational Jacobian polynomials;
- the 13-coordinate rational center and radius $r = 3 \times 10^{-14}$;
- exact rational enclosures for $\sin(k\pi/8)$, $k = 1, 2, 3$;
- a rational approximate inverse R ;
- rational center matrices N_c and $X_{3,c}$;
- declared entry Lipschitz constants and a Sylvester margin.

The checker imports only `fractions`, `json`, `pathlib`, `sys`, `typing`, and `__future__`. An abstract-syntax regression test enforces this boundary.

4.3 Exact Krawczyk computation

Let $F : \mathbb{R}^{13} \rightarrow \mathbb{R}^{13}$, let $X = x_c + [-r, r]^{13}$, and let $J(X)$ be the exact interval Jacobian. The verifier constructs

$$\mathcal{K}(X) = x_c - RF(x_c) + (I - RJ(X))(X - x_c). \quad (4.1)$$

All interval endpoints are `Fraction` objects. Strict coordinatewise inclusion

$$\mathcal{K}(X) \subset \text{int } X$$

is recomputed. The diagnostics are

$$\begin{aligned} \text{minimum inward margin} &\approx 1.8189522605 \times 10^{-14}, \\ \text{maximum contraction row sum} &\approx 6.8847446957 \times 10^{-10}, \\ \text{maximum point-residual radius} &\approx 6.6004906310 \times 10^{-17}. \end{aligned}$$

The exact value interval is

$$\frac{5233281002710117}{10^{16}} \leq L \leq \frac{5233281002710717}{10^{16}}, \quad |I_L| = \frac{3}{5 \times 10^{13}}. \quad (4.2)$$

4.4 Exact positivity payments

The checker performs exact fraction Gaussian elimination on every leading principal minor of

$$N_c - \frac{1}{100}I, \quad X_{3,c} - \frac{1}{100}I.$$

All are positive. Paying the full box perturbation gives

$$N \succeq \frac{24999999997}{2500000000000}I, \quad X_3 \succeq \frac{124999999997}{1250000000000}I. \quad (4.3)$$

4.5 Adversarial rejection

The test suite replaces the rational preconditioner by the zero matrix. Then (4.1) loses strict inward inclusion, and the checker returns failure. This demonstrates that the verifier does not merely echo a precomputed acceptance field.

4.6 O186 interface

O186 is the *Dependency-Minimal Exact O181 Replay Certificate*. Its domain is a finite rational JSON document; its codomain is an accept/reject decision with exact diagnostic margins. Removing monomials or the Jacobian prevents recomputation. Removing strict inclusion destroys the uniqueness certificate. Removing positivity payments leaves the operator root inadmissible for the attainment theorem.

O186 checks the finite root and positivity component. The Riccati-to-support and trace-telescope implications remain analytic theorems; O186 is not a physical measurement validator.

Chapter 5

P484: adversarial phase-face audit

5.1 Candidate complex tangent

Write $\Delta = iK$, where K is real skew. An imaginary Hermitian normalizer displacement has the form

$$N(t) = N + itQ, \quad Q^T = -Q.$$

Because (3.1) is linear in N , a flat affine direction must satisfy

$$X_3 Q X_3 + \frac{1}{4} K Q K = 0. \quad (5.1)$$

The P474 finite contact matrix had rank 127 in 128 real coordinates. An independent imaginary-causal tangent map had rank six in seven coordinates. Both calculations indicated one candidate direction, but neither was exact.

5.2 Exact parity reduction

An exact swap/complement basis gives the active sector

$$N = N_0 \oplus N_0, \text{quad} X_3 = X_+ \oplus X_-, \text{quad} K = \begin{pmatrix} 0 & C \\ -C^T & 0 \end{pmatrix}. \quad (5.2)$$

The executable proof checks nine exact representation identities: basis orthonormality, vanishing parity off-blocks, equality of the two normalizer blocks, the skew relation between the two K cross blocks, and both standard-sector formulas.

The positive Riccati congruences imply

$$X_- = \frac{1}{4} C^T X_+^{-1} C, \quad (5.3)$$

so with

$$B = \frac{1}{2} X_-^{-1} C^T$$

one obtains $BN_0B^T = N_0$. Conditional on $\det B = 1$, odd dimension gives a fixed covector and an invariant skew form. If this covector lies in the equal-endpoint causal plane, it lifts to a nonzero exact solution of (5.1). Exact positivity would then produce a nonzero constant-value segment.

5.3 The missing implication

The last sentence contains a genuine additional premise. Odd dimension does not force the fixed covector into the required causal plane.

Take

$$N_0 = X_+ = X_- = I_3, \quad v = (1, 0, 2)^T,$$

and define the rational half-turn

$$B = \frac{2vv^T}{v^Tv} - I_3 = \begin{pmatrix} -3/5 & 0 & 4/5 \\ 0 & -1 & 0 \\ 4/5 & 0 & 3/5 \end{pmatrix}, \quad C = 2B^T. \quad (5.4)$$

Exact symbolic arithmetic gives

$$BB^T = I_3, \quad \det B = 1, \quad B^Tv = v,$$

and

$$I_3 - CC^T/4 = 0, \quad I_3 - C^TC/4 = 0. \quad (5.5)$$

Thus both active Riccati blocks and orientation-preserving metric orthogonality hold, but $v_1 \neq v_3$. The claimed automatic causal-axis implication is [\[Refuted\]](#).

This counterexample does not satisfy all additional FIN shared-entry and standard-sector constraints. It therefore does not refute the specific P473 candidate. It proves exactly why those additional equations require a new identity or certificate.

5.4 Evidence that survives

At the P473 branch the computed map has

$$\det B \approx 1, \quad qquad\sigma(B) = \{1, 0.2704387824 \pm 0.9627371734i\}.$$

Its fixed covector, normalized by the first entry, is

$$(1, 0.02106502319, 0.9999999999999978)^T,$$

with equal-endpoint residual 2.22×10^{-15} . Six tangent equations return the common displayed value

$$\rho \approx 0.02979044149309996478959538442548707608969921737694 \dots$$

through 80 digits. A 180-digit rerun resolves differences near 10^{-102} , at the finite nonlinear-solve tolerance. The tangent residual is 6.65×10^{-16} .

These figures are strong numerical evidence. They cannot establish an exact zero or an exact affine face.

5.5 Corrected theorem status

The representation reduction and counterexample (5.4)–(5.5) are [\[Proven\]](#). The existence of an exact nonzero causal tangent at the exact P473 root is [\[Strong evidence\]](#) with one explicit missing identity. Full-cone optimizer uniqueness is neither proved nor refuted. O185 is therefore a *Conditional Odd-Dimensional Complex Comb Phase Face*.

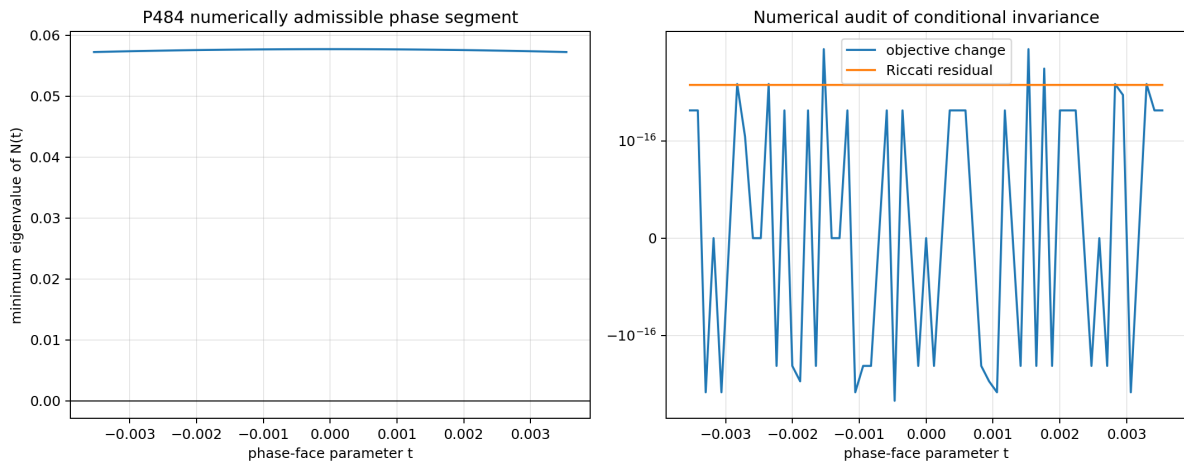


Figure 5.1: Numerical audit of the conditional P484 phase segment. Positivity and constant objective are sampled, while exact causality of the tangent remains the named proof obligation.

Chapter 6

P475: resource-bounded algebraic elimination

6.1 Exact algebraic coefficient field

Let $\alpha = \sin(\pi/8)$. Then

$$8\alpha^4 - 8\alpha^2 + 1 = 0, \tag{6.1}$$

and

$$\sin(\pi/4) = 1 - 2\alpha^2, \quad \sin(3\pi/8) = 3\alpha - 4\alpha^3. \tag{6.2}$$

The extension degree is four. Indeed, for $y = 2\alpha$, $y^4 - 4y^2 + 2$ is Eisenstein at 2.

After (6.2), the exact problem contains 13 equations in 13 operator coordinates over $\mathbb{Q}(\alpha)$. Each equation has operator-coordinate degree at most three. Representing α as a fourteenth rational-elimination variable raises the largest total degree to seven. The largest equation has 29 monomials, and the full list has 262 monomial occurrences.

6.2 Affine normalizer structure

Every Riccati equation is affine in A, B, u , the three normalizer coordinates. A floating pivot scout selects equation rows 5, 10, and 11. The corresponding exact determinant is a nonzero polynomial of total degree 10 with 409 terms; at the floating root its absolute value is approximately 0.02238991248.

This is a route proposal, not yet a certified elimination: nonzero as a polynomial and nonzero at a floating locator do not prove nonvanishing over the exact root box.

6.3 Declared elimination envelope

The exact worker adds (6.1) and requests a lexicographic Gröbner basis in 14 variables, ordering L last. It is restricted to

- 40 seconds CPU time;
- 45 seconds wall time;
- 3 GiB address space;
- SymPy 1.12 on the local Intel i3-10110U / 16 GB host;
- no network, external CAS, or remote computation.

The parent terminated the worker after 45.0778 seconds. No univariate relation in L was returned. The crude number

$$3^{13} = 1,594,323$$

is only a projective cubic Bézout bound over $\mathbb{Q}(\alpha)$; it is not a root count and does not incorporate positivity or local branch admission.

6.4 Interpretation of the no-go

P475 proves the exact coefficient-field reduction, polynomial inventory, affine normalizer dependence, and faithful resource stop. It does not prove transcendence, nonexistence of a minimal polynomial, ideal dimension, algebraic degree, or global solution count.

O187 is the *Resource-Bounded Algebraic Elimination Burden Certificate*. It can be falsified constructively by a future modular, resultant, homotopy-assisted, or structure-aware exact workflow that returns and verifies a relation under a comparable envelope.

Chapter 7

Falsification ledger

Candidate claim	Attack	Verdict
P480 merely trusts a stored verdict	Zero every preconditioner entry	Rejected; strict inclusion fails
P480 needs the scientific Python stack to verify	Audit imports and replay from JSON	Refuted; standard library suffices
Odd dimension automatically supplies the causal P484 tangent	Exact rational half-turn with axis $(1, 0, 2)$	Refuted for active Riccati/metric premises alone
The P473 phase candidate is merely numerical noise	Independent contact and tangent maps; precision escalation	Strong evidence against noise, but exact zero still open
Direct exact lexicographic elimination is cheap locally	Hard-bounded 14-variable run	Refuted only inside the declared implementation/resource envelope
P475 proves no minimal polynomial exists	Logical boundary audit	Refuted; no such conclusion follows from timeout

Chapter 8

Kernel, selector, dimensional, and physical audit

The binding split remains

$$K_{\text{legacy,ont}}(d) = \frac{\alpha_{\text{geo}} \cos(\omega d + \phi)}{1 + \beta_{\text{tors}} d}, \quad K_{\text{strict,gate}}(d) = \frac{\cos(\omega d + \phi)}{1 + \beta d^\eta}. \quad (8.1)$$

Programs P475/P480/P484 analyze a supplied downstream finite channel. They derive neither member of (8.1), complete no bridge, and transfer no legacy electroweak, electromagnetic, gravitational-hierarchy, or other physical role.

QW-2191 remains open. A complex affine degree of freedom, even if P485 later proves it exact, is not automatically a selector, orientation source, or observable. It supplies no clock, preparation, instrument, environment, apparatus, observer, or measurement record.

All quantities remain dimensionless. Mathematical duality

$$U_t = e^{-itA}, \quad P_t = e^{-tA}$$

does not generate a calibrated physical time or laboratory implementation. No synthetic datum is treated as empirical evidence.

The internal ontology remains

$$\text{nadsoliton} \longrightarrow \text{light} \longrightarrow \text{matter} \longrightarrow \text{emergent observer},$$

with no informational layer beneath the nadsoliton. This checkpoint neither falsifies nor physically validates that ontology.

Chapter 9

Recommended next programs

Rank	Program	Decisive local question	Chance
1	P485	Exact ideal identity or certified counterexample for the P473 shared-entry/standard-sector causal-axis implication	55%
2	P486	Certify $\det C > 0$, $\det B = 1$, and all orientation premises uniformly on the P473 box	80%
3	P487	Eliminate A, B, u through a certified pivot and try a ten-variable structure-aware resultant/Gröbner route	50%
4	P488	Finite-field Gröbner scouting plus Chinese-remainder reconstruction of a candidate polynomial for L	35%
5	P489	Conditional on P485, certify persistence of the complex phase face on a nonzero P483 sub-tube	30%
6	P490	Build an independent second P480 implementation and differential-test exact verdicts	75%
7	P491	Bound the local algebraic degree of O181 without computing the full minimal polynomial	45%
8	P492	Certify the complete complex contact-face dimension or a transverse rank lower bound	40%
9	P493	Interval-certified sensitivities dL/dq and $dL/d\theta$	75%
10	P494	Grow the P483 parameter cover until a certified inclusion, positivity, or conditioning boundary	70%
11	P495	Seek a bounded four-slot support theorem or a counterexample from the exact two-/three-slot reductions	30%
12	P496	Formalize the minimum operational data needed to make O167 observable and prove the local empirical boundary	85%
13	P497	Audit which O167 inputs are kernel-split robust and which require a typed completion atom	70%

The selected next bounded batch is

$$\boxed{P485 \longrightarrow P486 \longrightarrow P487}. \quad (9.1)$$

P485 decides whether P474/P484 becomes an exact nonuniqueness theorem or remains numerical evidence. P486 pays the orientation premise separately. P487 exploits P475's affine structure instead of repeating blind full lexicographic elimination.

Chapter 10

Reproducibility and artifact hand-off

10.1 Restart commands

```
MPLCONFIGDIR=/tmp/matplotlib python3 fin_program_484.py
MPLCONFIGDIR=/tmp/matplotlib python3 fin_program_480_build_certificate.py
python3 fin_program_480_standalone_checker.py
MPLCONFIGDIR=/tmp/matplotlib python3 fin_program_475.py
python3 -m unittest -v test_fin_programs_480_484.py
```

P475 intentionally consumes up to 45 seconds and is expected to report the bounded timeout on the present host. P480 replay itself requires no scientific Python dependency.

10.2 Primary artifacts

- FIN_Local_Research_Checkpoint_P475_P480_P484_EN.md
- FIN_Local_Research_Checkpoint_P475_P480_P484_EN.tex
- FIN_Local_Research_Checkpoint_P475_P480_P484_EN.pdf
- FIN_Local_Research_Checkpoint_P475_P480_P484_State.json
- FIN_Program_475_Results.json
- FIN_Program_475_Algebraic_Inventory.csv
- fin_program_475.py
- fin_program_475_elimination_worker.py
- FIN_Program_480_Standalone_Certificate.json
- FIN_Program_480_Standalone_Check_Result.json
- fin_program_480_build_certificate.py
- fin_program_480_standalone_checker.py
- FIN_Program_484_Results.json
- FIN_Program_484_Symmetry_Phase_Face.csv
- FIN_Program_484_Phase_Face_Witness.npz
- FIN_Program_484_Figures/p484_exact_complex_phase_face.png
- fin_program_484.py
- test_fin_programs_480_484.py
- RELEASE_10_46_PROGRAMS_475_480_484.md
- FIN_Checkpoint_P475_P480_P484_AGENTS_Guardrail.txt
- FIN_PROGRAMS_475_480_484_RELEASE_MANIFEST.sha256

Chapter 11

Conclusion

Release 10.46 improves rigor in two opposite but complementary ways. It makes the exact O181 certificate substantially easier to audit, and it withdraws an exact phase-face conclusion that the current artifacts do not yet prove.

The mathematical frontier is now sharp. Either the additional FIN shared-entry and standard-sector equations force the observed causal axis exactly, or the near-perfect numerical phase direction is only approximate. P485 can ask this as a finite, falsifiable local problem. No cosmological or physical interpretation is needed to state or decide it.

No selector closure, dimensional source, laboratory validation, legacy-to-strict completion, role transfer, L_{total} , Standard Model, gravity, or Theory-of-Everything closure is claimed.